

Jatin Shukla

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Professional Summary

Senior AI/ML Engineer with **2.8+ years of total experience** building production-grade **AI agents, RAG systems, machine learning models, and full-stack AI applications**. Strong expertise in **Python, SQL, FastAPI, PyTorch, Scikit-learn, LangChain, semantic retrieval, prompt engineering, vector search, and cloud deployment**. Proven ability to translate business problems into scalable AI solutions, improve model quality and latency, and deliver measurable impact through user-facing AI products and intelligent automation.

Technical Skills

Languages: Python, SQL, JavaScript, C++, C#

AI / ML: Machine Learning, Deep Learning, PyTorch, Scikit-learn, Regression, Classification, Clustering, Feature Engineering, Model Evaluation, EDA, Experimentation

GenAI / Agents: AI Agents, LLM Applications, RAG, Prompt Engineering, LangChain, Semantic Search, Context Grounding, Information Retrieval, Agentic Workflows

Data / Retrieval: FAISS, Vector Search, BM25, Cross-Encoder Reranking, Document Parsing, Unstructured Data Processing, TF-IDF, Cosine Similarity, Sentence Transformers, Resume & JD Parsing

Frameworks / Deployment: FastAPI, Flask, React, Vite, Tailwind CSS, REST APIs, Docker, Azure, CI/CD, Vercel, Render, Joblib, Git, GitHub

Experience

Senior Engineer – AI/ML

Mar 2026 – Present

Havells India Ltd., Noida

- Built **AI agents for applicant shortlisting** for HR workflows, automating resume screening, JD matching, and ranking logic to reduce manual review effort by **40%+** and accelerate shortlist turnaround.
- Developed **agentic ideation, research, and prototyping workflows** for internal use, enabling structured **10-step reasoning pipelines** for faster concept exploration and improving early-stage solution turnaround by **30%**.
- Enhanced internal AI products with **retrieval, prompt orchestration, scoring logic, and user-facing workflows**, improving usability and consistency for business teams across model-assisted decision tasks.

AI/ML Engineer

Jan 2025 – Feb 2026

Havells India Ltd., Noida

- Built and deployed **ML models for the fan team** to predict **air delivery, velocity, and pressure on fan blades**, improving engineering prediction accuracy by **18–22%** and reducing iterative validation effort.
- Developed **ML models for the LED team** to estimate **heat generation at critical points**, supporting thermal-performance analysis and reducing design evaluation turnaround by **35%**.
- Engineered a production inference pipeline for a **Tri-Branch PointNet regression model**, reducing end-to-end latency by **35%** through optimized preprocessing, feature scaling, and joblib-based deployment.
- Designed automated **data preprocessing, feature engineering, and inference consistency pipelines** using **Python, PyTorch, Scikit-learn, and joblib**, improving model reliability across environments.
- Built **UI-driven internal applications** for model interaction using **Flask, React, and VTK.js**, enabling **30+ engineers** to use AI outputs through business-friendly interfaces and cross-platform packages (**.exe, .deb**).
- Created scalable **3D / point-cloud evaluation workflows** handling **10+ design configurations per week**, reducing manual analysis and dependence on physical prototyping.

Projects

Nexora – AI Study Assistant

Personal Project — [Live](#) — [GitHub](#)

- Built **Nexora**, a full-stack **RAG-based AI knowledge assistant** using **React, Vite, FastAPI, and Supabase** to manage app data and support grounded Q&A over **PDFs and web URLs**.
- Engineered a **hybrid retrieval pipeline** with **FAISS** for dense semantic search, **BM25** for sparse retrieval, and **Cross-Encoder reranking**, improving relevance over vector-only search.
- Developed an AI learning workflow with **quiz generation, scoring, explanations, weak-topic feedback, and 6 answer modes**, turning static content into an interactive study system.

HireFit – AI Resume–Job Matcher

Personal Project — [Live](#) — [GitHub](#)

- Built a full-stack **AI/ML and GenAI application** using **FastAPI, React, Vite, Tailwind CSS, Python, and Gemini API** for resume-JD matching, ATS scoring, semantic comparison, and recruiter-facing recommendations.
- Designed **structured evaluation pipelines** combining TF-IDF, cosine similarity, evidence validation, prompt-based reasoning, and multi-factor scoring for accurate profile-to-role assessment.
- Implemented features including keyword coverage analysis, shortlist-risk analysis, safer resume rewrites, multi-JD comparison, and export-ready outputs for end users.

IoT-Enabled Digital Twin for Post-Harvest Loss Minimization

Research Project — [Link](#)

- Built an **Azure-based AI + IoT system** using Raspberry Pi and sensors for predictive analytics, real-time monitoring, and decision support, reducing potential post-harvest losses by **15–20%**.
- Integrated cloud-connected sensing, analytics, and predictive workflows into a deployable digital twin architecture for storage optimization and operational visibility.

Education

National Institute of Technology, Hamirpur

2023 – 2025

M.Tech, Computer Science and Engineering

CGPA: 9.15/10

Institute of Engineering and Technology, Lucknow

2018 – 2022

B.Tech, Chemical Engineering

CGPA: 7.8/10